

ANOVA test:

Q1 — What does ANOVA stand for? What is its primary purpose?

ANOVA stands for **Analysis of Variance**.

The primary purpose of ANOVA is to determine whether there are **statistically significant differences between the means of three or more independent groups**. Instead of comparing groups individually, ANOVA evaluates overall variation to identify meaningful differences.

Q2 — How is ANOVA different from a t-test?

Aspect	t-test	ANOVA
Groups compared	2 groups	3 or more groups
Purpose	Compare two means	Compare multiple means
Error control	Higher risk with repetition	Controls Type I error
Efficiency	Simple comparison	Global comparison method

A t-test is suitable for two groups, while ANOVA is designed for multiple group comparisons.

Q3 — Real-world examples of ANOVA

ANOVA is used when comparing multiple conditions, such as:

- Comparing student performance under different teaching methods
 - Testing drug effectiveness across dosage groups
 - Evaluating productivity under different work schedules
 - Comparing marketing campaign performance
 - Measuring crop yields under various fertilizers
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Q4 — Null hypothesis (H_0) in ANOVA

The null hypothesis states:

All group means are equal.

Mathematically:

$$H_0: \mu_1 = \mu_2 = \mu_3 = \dots = \mu_k$$

This assumes no meaningful difference exists between groups.

Q5 — Alternative hypothesis (H_1) in ANOVA

The alternative hypothesis states:

At least one group mean is different.

ANOVA does not specify which group differs — only that a difference exists.

Q6 — Basic assumptions of ANOVA

ANOVA relies on:

- ✓ Independence of observations
- ✓ Normally distributed populations
- ✓ Equal variances (homogeneity of variance)

Violating these assumptions may invalidate results.

Q7 — One-Way vs Two-Way ANOVA

One-Way ANOVA

- Compares groups based on **one independent factor**
- Example: Comparing test scores across teaching methods

Two-Way ANOVA

- Evaluates **two independent factors**
 - Also checks interaction effects
 - Example: Teaching method + study time impact on scores
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Q8 — Between-group vs Within-group variance

Between-group variance

Measures variation **due to differences between group means**.

Within-group variance

Measures variation **inside each group**, caused by natural randomness.

ANOVA compares these to detect meaningful differences.

Q9 — F-statistic in ANOVA

The F-statistic is calculated as:

$$F = \text{Between-group variance} / \text{Within-group variance}$$

Interpretation

- Large F-value → Strong evidence of group differences
 - Small F-value → Differences likely due to chance
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Q10 — Meaning of p-value < 0.05

If:

$$p\text{-value} < 0.05$$

We reject the null hypothesis.

This means there is **statistical evidence** that group means differ.

Q11 — Degrees of freedom (df)

Degrees of freedom measure independent variability:

$$\begin{aligned} df_{\text{between}} &= k - 1 \\ df_{\text{within}} &= N - k \end{aligned}$$

Where:

- k = number of groups

- N = total observations
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Q12 — Action after significant ANOVA result

If ANOVA is significant:

 Perform **post-hoc tests** to identify which groups differ.

Examples:

- Tukey's test
 - Bonferroni correction
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Q13 — Why post-hoc testing is needed

ANOVA only indicates **that** differences exist.

Post-hoc tests determine:

- ✓ Which specific groups differ
 - ✓ Magnitude of difference
 - ✓ Error control
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Q14 — Risk of multiple t-tests

Running many t-tests increases **Type I error**:

False positives become more likely.

ANOVA prevents this by performing a single global comparison.

Q15 — When should ANOVA not be used?

Avoid ANOVA when:

- ✗ Data is not normally distributed
- ✗ Variances are unequal
- ✗ Samples are dependent

- ✗ Very small sample sizes
- ✗ Non-numeric data

Alternative non-parametric tests may be better.

Summary

ANOVA is a powerful statistical tool for comparing multiple groups efficiently while controlling error rates. Understanding assumptions, interpretation, and follow-up testing is essential for correct analysis.